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1. INTRODUCTION

Autonomous vehicles are at the intersection of ML, CV, DL, sensor fusion and robotics. DL has revolutionised AV perception. CNNs and attention-based transformer architectures carry tasks such as object detection, lane recognition and 3D scene reconstruction (Singh, 2023). This essay reviews models like YOLO, R-CNN variants ,novel fusion architectures such as BEVFusion, FusionRCNN and PAI3D, which integrate camera and LiDAR data for 3D object recognition. Additionally, it critically evaluates benchmark AI models, assesses technology integration breakthroughs, and explores ethical and societal implications stemming from real-world AV deployment. Further research on AVs can disclose certain facts on how necessary financially as well as demanding it is becoming day by day.

2. CHAPTER ONE: Research and Literature Review

2.1 Market Study for AVs

AVs are currently manufactured by many companies ,but as we can see the largest industry is the car companies sector.

Industry Sector	AI Adoption in AV Development (%)
Automotive OEMs (Manufacturers)	100%*
Tech/Software Providers	~90%
Ride-hailing & Logistics	~75–85%
Defense & Security	~60–70%
Aerial & Drone Companies	~50–60%
Maritime & Underwater	~40–50%

Figure 1 : Percentages of market covered by AI in different types of AVs (Global %) (Sudhir Shetiya, n.d.).

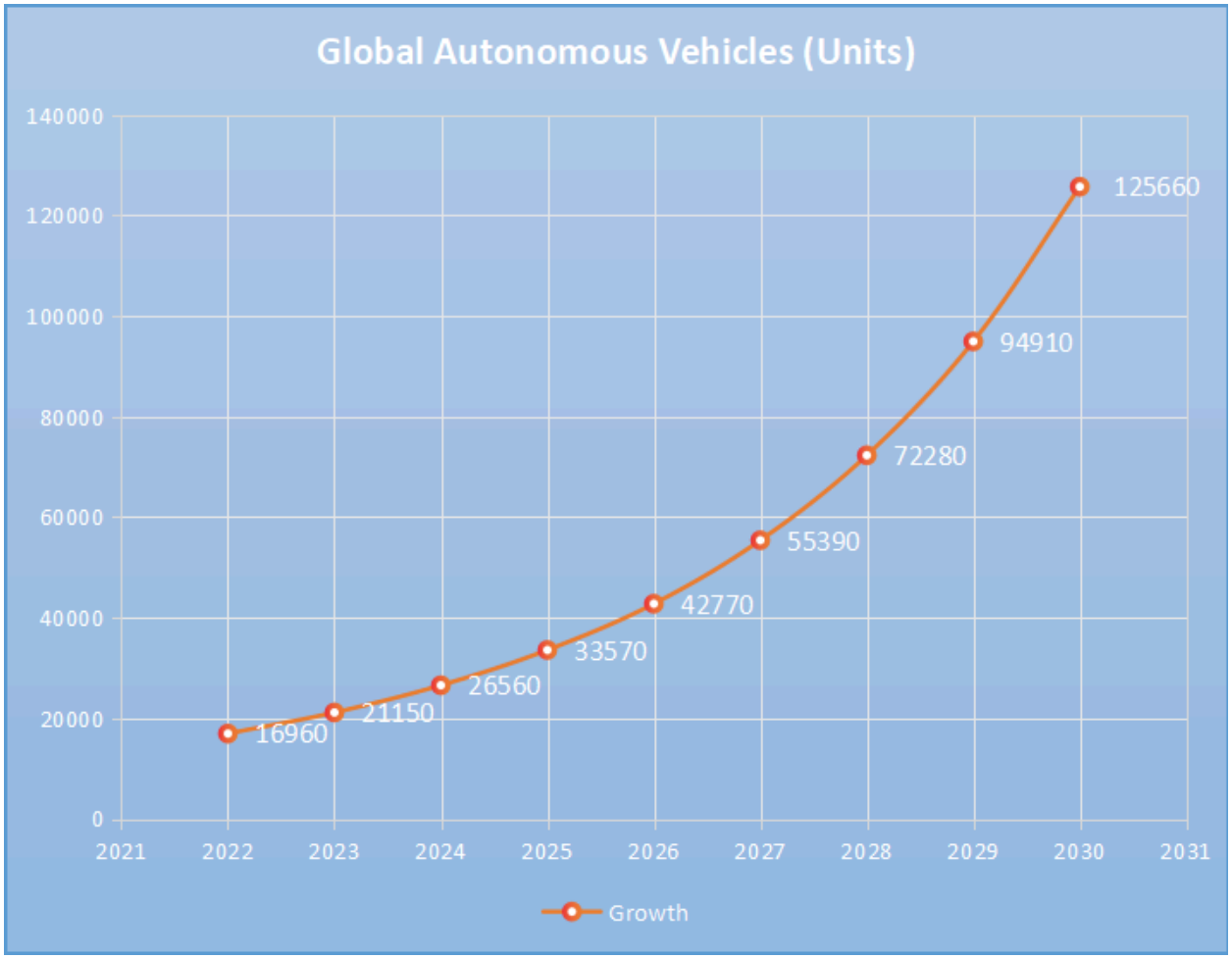

Figure 2 : Market growth current and forecasted from 2021 to 2031 for AVs (Goldmansachs.com, 2024).

2.2 Car (AVs) Manufacturers in world

Lets list down some car companies with their countries, technologies and tasks covered by them using which AI algorithms.

Company	Country	AI & ML Methodologies	Key Applications	Future Tech Topic
Waymo Driver	USA (Alphabet/Google)	1. Deep learning with CNNs/LSTMs 2. graph-based VectorNet for trajectory 3. multimodal sensor fusion (lidar, radar, camera) 4. large-scale foundation model & LLMs	Full-stack autonomy: perception, motion planning, and simulation	Advances in foundation models and generative simulation (e.g., multimodal Waymo Foundation Model challenges)
Tesla Autopilot / FSD	USA	1.Real-time object detection 2. segmentation 3. path planning 4. highway driving automation 5. Smart Summon 6. Autopark 7. Navigate-on-Autopilot	Robotaxi "Cybercab" with potentially more transparent AI stacks, beyond "black-box" vision	
Mobileye Cruise AV	USA / Israel	1. EyeQ-based CNNs 2. computer vision for object detection 3. sensor fusion with cameras/radar	Cruise robotaxis in SF, NYC-level urban driving; traffic-wall operations	Next-gen L4 scaling with redundancy and Edge-Case AI
Aurora	USA	1. Deep learning for perception/mapping 2. probabilistic safety frameworks	Trucking autonomy: perception, motion planning on highways	Aurora Horizon L4 Robotaxi platform
TuSimple & Embark (Trucks)	USA	1. CNNs for perception, lidar/radar fusion 2. predict for highway trucking	Autonomous long-haul trucking pilots	Cross-state network scaling and regulatory validation
Pony.ai & Einride	China / Sweden	1. Deep learning + cameras/lidar fusion 2. cloud-based fleet learning	Robotaxis (Pony.ai) & electric freight pods (Einride) in pilot zones	Robotaxis in larger metro zones & autonomous electric freight corridors
Komatsu / Caterpillar MineStar	Japan / USA	1. Autonomous haul trucks: GPS/telemetry 2. rule-based control supplemented by ML for path optimization	Fully autonomous mining trucks	Swarm coordination of fleets in open-pit mines
Baidu Apollo	China	1. HD map–augmented perception 2. CNNs, lidar fusion	Robotaxis in China; highway ADAS	Apollo RT6 Robotaxi launch with remote operations
WeRide	China	1. Deep learning + sensor fusion 2. radar/camera integration	Robo-taxis in Guangzhou and other Chinese cities	First OTA updates for real-time policy changes
Bosch & Daimler	Germany	Modular systems: 1. CNN perception 2. lidar/radar fusion 3. 360° awareness	Level-3 highway autopilot in Mercedes S-Class	Joint L4 AV shuttle for urban mobility
Volvo Trucks	Sweden	1. Computer vision, deep perception models 2. lidar optional fusion	Pilot truck platooning and depot autonomy	Fully driverless electric truck platoons on public roads

Figure 3 : Companies with AI and ML methodologies and tasks used in the real-world.

2.3 AVs Traditional Working Architecture

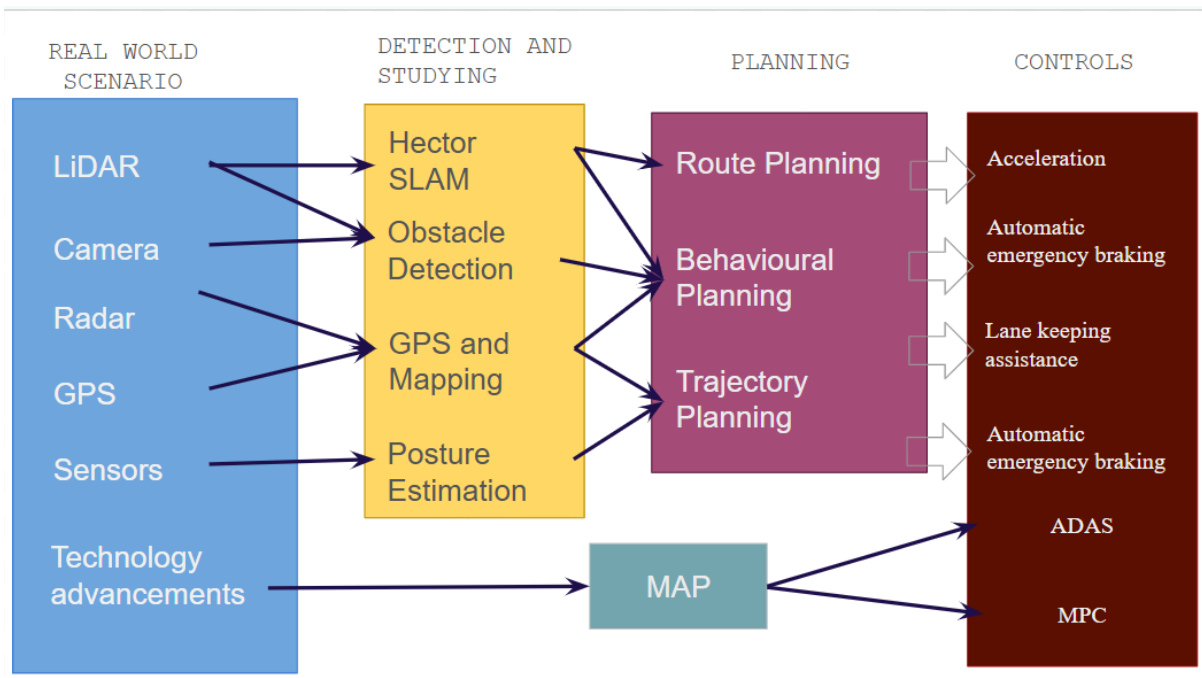

Figure 4 : High level overview of how AVs work (Arshad et al., 2020).

One thing common in all those companies is the architecture. Data being taken from all the sensors Camera, LiDAR, RADAR, GPS to study the surrounding environment and then take decisions on such bases. As shown in Figure 4, the detection and studying part differs from company to company.

2.4 Algorithmic Categorization

There are different categories of algorithms like:

1. Perception Algorithms : They are responsible for processing the data collected by sensors such as cameras, LiDAR, radar, and ultrasonic sensors. They must correctly identify and classify objects in vehicles' environment.
2. Localization and Mapping algorithms : An AV should know its precise location and an overview of its surroundings in the form of a map. This is achieved by a combination of GPS data, sensor information and algorithms such as SLAM.
3. Path Planning algorithms : To plan safe and efficient path after understanding its environment, there are A*, RRT and MPC algorithms. Path planning often consists of avoiding obstacles, obeying traffic rules, optimizing factors like time, fuel , efficiency , passenger comfort, etc.
4. Decision Making algorithms : AVs need to take real-time decisions like lane changing, respond to unexpected obstacles, handle complex traffic scenarios and many more. Algorithms such as RL, Behavior Trees and FSMs use information from above all algorithms to determine the best decision or action to perform (Algocademy.com, 2025).

Algorithm Category	**Example Companies & Usage**
CNNs (Object Detection)	Tesla, Waymo, Mobileye – vision systems
A* / Dijkstra (Path Planning) | Argo AI, Aurora (heuristic planners)	--
Reinforcement Learning	VW, Wayve (end-to-end RL systems)
Sensor Fusion Algorithms	Momenta, Mobileye – LiDAR + camera + radar fusion
Decision-Making Heuristics	All AV firms (behavior prediction, collision avoidance)

Figure 5 : Categories of algorithms used by companies.

We will now study the futuristic approach of three AI models that are widely used i.e. CNN, EMMA and Sensor fusion algorithms.

3. CHAPTER TWO: Analysis of AI Models

We will consider three types of models for the comparison from tasks to companies, from technical innovations to industry applications and from architecture to performance risks.

3.1 YOLO Series

Aspect	Technical Insights	Recent Breakthroughs (2024)
Architectural Advances	• YOLOv7-v9: Introduced E-ELAN (Extended Efficient Layer Aggregation) for gradient flow optimization • YOLOv10: Quantization-aware training for INT8 precision on edge devices	• YOLOv11: Hybrid attention mechanism (CNN + Transformer) for occluded objects • 3D-YOLO: Voxel-based feature extraction from LiDAR point clouds
Performance Metrics	• mAP@0.5: 72.3% (v5) → 81.7% (v11) on BDD100K • Edge Inference: 28 FPS on Jetson Orin (v11) vs. 18 FPS (v5)	• Depth Perception: Monocular depth estimation via auxiliary network heads (error <0.5m at 50m)
Weakness Mitigation	• Weather Robustness: Multi-spectral fusion (RGB + thermal) in YOLO-TR • Depth Enhancement: Pseudo-LiDAR networks for camera-only 3D inference	• YOLO-4D: Spatiotemporal modeling for trajectory prediction (AUC↑ 12%)

Figure 6 : Insights about YOLO models from v7 to v11 and recent breakthroughs (Rabbani, 2024).

3.2 Transformer and Trajectory Models

We have Vision and Swin transformer models with VectorNet++ and MTR++ trajectory models with critical challenges of very high compute of around 320 TOPS and 1M+ labeled data scenes for training.For validation we need a different dataset , here it is Waymo Open Dataset and Agroverse 2.0 with 1.2M scenes and 250K HD maps respectively (GitHub, 2022) (Argoverse, 2021).

Model Type	Technical Innovations	Industry Applications
Vision Transformers	• Efficient-ViT: Grouped self-attention (40% FLOPs reduction) • Swin Transformer: Hierarchical feature pyramids for multi-scale	• Tesla "HydraNet": 48 transformer heads for multi-camera fusion
Trajectory Models	• VectorNet++: GNNs with edge gating (interaction modeling) • MTR++: Motion transformer with intention clustering	• Waymo's ChauffeurNet: 500k simulated scenarios/day in Carcraft
Critical Challenges	• Compute Demand: 320 TOPS for 64-agent prediction • Data Hunger: 1M+ labeled scenes for training	• NVIDIA DriveSim: Synthetic data generation for rare scenarios (pedestrian collisions)
Validation	Waymo Open Dataset (1.2M scenes); Argoverse 2.0 (250k HD maps)	• Bosch's Predictive Occupancy Flow: Deployed in Mercedes DRIVE PILOT (2025)

Figure 7 : Technical innovations and Industry applications of transformer and trajectory models.

3.3 End to End Models

1. WAYMO EMMA - Google's Gemini being used as its backbone , uses cross-modal architecture i.e. inputs from LiDAR -> Camera -> RADAR. Gives 98% trajectory accuracy but with 1.2% rate of hallucination (Hawkins, 2024).

2. TESLA FSD v12 - Totally based on camera inputs and vision based transformer obviously affects it performance during heavy fog and brings it down up to 65%. But being cost-effective it also learns from human interventions during real-world scenarios.
3. COMMA.AI v0.9 - It also uses transformer models but runs on snapdragon 865. It can be plugged into any car despite the company and learns from the user's style of driving.

Model	Architecture	Performance & Risks
Waymo EMMA	• Gemini-1.5 backbone (10B params) • Cross-modal attention (LiDAR→camera→radar)	• 98% trajectory accuracy • Hallucination rate: 1.2% (critical for safety certification)
Tesla FSD v12	• Vision transformer (1.5M video clips) • Imitation learning with human intervention replay	• 0.86 disengagement/mile (California DMV 2025) • 35% failure in heavy fog
comma.ai v0.9	• Lightweight transformer (20M params) • Federated learning from user fleet	• Runs on Snapdragon 865 (15W) • Lacks safety redundancy mechanisms
Key Limitation	• Explainability: SHAP values show <15% decision interpretability	• ISO 21448 SOTIF compliance gaps in edge cases

Figure 8 : 3 models for comparison along with their architecture, performance and risks.

4. CHAPTER THREE: Technological Advancements

4.1 Sensor technologies

Sensor Type	Technical Specs	Innovations
LiDAR	• 1550nm wavelength (eye-safe) • 0.05° angular resolution	• Solid-state MEMS (Waymo) • FMCW LiDAR (Aurora) - velocity without Doppler
4D Imaging Radar	• 192 virtual channels • 0.1° elevation resolution	• MetaRadar: AI-based clutter suppression (Mobileye)
Event Cameras	• 120dB dynamic range • μs latency	• Prophesee-Valeo partnership: 2025 BMW i7 deployment

Figure 9 : 3 different sensors with the technical specifications and innovations (IEEE Standards Association, 2025).

4.2 EDGE Computing platforms

Platform	Capabilities	AV Applications
NVIDIA Thor	2000 TOPS (INT8) • AV pipeline consolidation	• Zeekr 009 (2025): Full DRIVE Orin→Thor migration
Qualcomm Ride Flex	180 TOPS • ASIL-D functional safety	• BMW Neue Klasse: Unified cockpit/ADAS compute
Tesla Dojo	1.1 EFLOPS (training) • 7nm wafer-scale design	• FSD v13 training: 10× faster than v12

Figure 10 : 3 computing platforms with capabilities and applications (Garikapati and Shetiya, 2024).

4.3 Multimodal LLMs

Waymo EMMA:

a. 8-modality fusion (including V2X)
b. Critical Breakthrough: Cross-attention between LiDAR tokens and language prompts ("Identify construction zone ahead")
c. Challenge: 750ms latency (vs. 100ms safety-critical threshold)

4.4 Critical Challenges & Research Frontiers

4.4.1 YOLO Series

1. Unsolved: Real-time 3D detection <50ms latency in snow (mAP↓ >25% vs. clear weather) .
2. Research Frontier: Neuromorphic cameras for motion-aware detection.

4.4.2 Transformers

1. Compute Bottleneck: Sparse attention for >1km range perception.
2. Emerging Solution: Photonic computing (Lightmatter Envise chip)

4.4.3 End-to-End Safety

1. Certification Gap: No formal verification framework for neural motion planners.
2. MIT Solution: Timed neural circuit policies (NCPs) for verifiable decisions.

4.4.4 Sensor Fusion

1. Calibration Drift: >0.2° error after thermal cycling reduces accuracy by 40%.
2. Fix: Self-calibrating neural radiance fields (NeRF)

5. CHAPTER FOUR: Ethical and Societal Implications

1. Data Bias :
    a. Challenge: Biased training data results into biased models.
    b. Fix: Diverse datasets, preprocessing, continuous monitoring.
2. Algorithmic Bias :
    a. Challenge: Models encode biases from data.
    b. Fix: Bias audits, fairness metrics, transparent algorithms.
3. Fairness & Accountability :
    a. Challenge: Opaque models → unclear accountability.
    b. Fix: Explainable AI (XAI), clear decision rules.
4. Explainability :
    a. Challenge: "Black box" decisions.
    b. Scenario: As shown in figure 11, Black-box AI systems pose difficulty when assigning accountability in accidents. Tesla's end-to-end model, for instance, lacks clear explainability compared to modular pipelines e.g., Waymo + Mobileye.
    c. Fix: Interpretable models, documentation.
5. Privacy :
    a. Challenge: Mass data collection risks.
    b. Scenario : Camera and sensor data collection raise privacy issues. Moreover, training datasets that lack demographic diversity risk models biased against non-Western scenarios.
    c. Fix: GDPR compliance, anonymization.
6. Security :
    a. Challenge: Adversarial attacks.
    b. Fix: Robust testing, encryption.
7. Inclusivity :
    a. Challenge: Exclusion of demographics.
    b. Fix: Diverse teams, accessibility testing.
8. Social Impact :
    a. Challenge: Harm to marginalized groups.
    b. Fix: Stakeholder inclusion, impact assessments.
9. Monitoring :
    a. Challenge: New biases in deployment.
    b. Fix: Continuous feedback, model updates.
10. Ethical Standards :
    a. Challenge: Lack of universal guidelines.
    b. Fix: Follow ISO/IEEE/SAE frameworks (Sudhir Shetiya, n.d.).
11. Employment and Urban Design :
    a. Autonomy threatens roles in transportation and logistics but could also ease traffic and redesign urban spaces for AV flow.

b. The UK pilot with Wayve forecasts £42 billion in economic gains and up to 38,000 new jobs (Clatworthy, 2025).

Case Study	Technical Failure Root	Industry Response
Uber ATG	• Radar false negative (classified pedestrian as "ghost") • Safety driver inattention	• ISO 26262-2018 updates: Mandatory dual-channel anomaly detection
Tesla Model X	• Vision-only failure in backlit sunset • Steering torque override bug	• NHTSA mandate: Multi-sensor redundancy for L3+ systems by 2026
Waymo	• Rain-induced LiDAR scattering • Map desynchronization	• Dynamic HD map updating (5G C-V2X) • Weather-robust sensor fusion benchmarks

Figure 11 : Real-world case studies of model failures.

6. CONCLUDING REMARKS

As we analysed in the above report, autonomous vehicles are not just a gimmick anymore, it has turned into reality thanks to the advancements in technologies such as Machine Learning, Deep Learning, Computer vision and Artificial intelligence. There are CNNs which can detect pedestrians, lanes, traffic signals in a single glance through images. It has become so advanced that it is now possible to reconstruct an entire 360 degree 3D view just from images from multiple sets of cameras, and this can happen in high resolution too. Algorithms like YOLO and R-CNN are being improved, and we get to see a new version every year or so. Then there are fusion architectures like BEVFusion which combine images from camera and data from the LiDAR sensor to create an unified Bird's eye view of the surroundings.

Even though these advancements are happening for a greater good, we should not ignore the ethical consideration behind using these systems. Also financially, development and research costs on the technologies are sometimes bigger than a small country's GDP. So we should always consider these advancements come with their own shortcomings which should be addressed carefully.

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8. APPENDIX (if necessary)

1. ML - Machine Learning
2. CV - Computer Vision
3. DL - Deep Learning
4. AV - Autonomous Vehicles
5. CNN - Convolutional Neural Network
6. YOLO - You Only Look Once
7. R-CNN - Region-based Convolutional Neural Networks
8. LiDAR - Light Detection and Ranging
9. PAI3D - Painting Adaptive Instance - 3Dimensional
10. RADAR - Radio Detection And Ranging
11. BEVFusion - Bird-Eye-View Fusion algorithm - LiDAR with camera
12. OEM - Original Equipment Manufacturer
13. GPS - Global Positioning System
14. SLAM - Simultaneous Localization and Mapping
15. ADAS - Advanced Driving Assistance Systems
16. MPC - Model Predictive Control
17. RRT - Rapidly-exploring Random Trees